

```

ls
teja@LAPTOP-FGDPB110:~$ ls
OSSP  Wsl3  process  process.c  process.c.save  process.c.save.1
teja@LAPTOP-FGDPB110:~$ pwd
/home/teja
teja@LAPTOP-FGDPB110:~$ data
Command 'data' not found, did you mean:
  command 'date' from deb coreutils (9.4-3ubuntu6.2)
  command 'dat' from deb liballegro4-dev (2:4.4.3.1-4)
Try: sudo apt install <deb name>
teja@LAPTOP-FGDPB110:~$ whoami
teja
teja@LAPTOP-FGDPB110:~$ uname
Linux
teja@LAPTOP-FGDPB110:~$ uname -r
6.18.33.2-microsoft-standard-WSL2
teja@LAPTOP-FGDPB110:~$ id
uid=1000(teja) gid=1000(teja) groups=1000(teja),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev),100(users),107(netd
ev)
teja@LAPTOP-FGDPB110:~$ tty
/dev/pts/0
teja@LAPTOP-FGDPB110:~$ uptime
 05:55:35 up 2 min,  1 user,  load average: 0.02, 0.02, 0.00
teja@LAPTOP-FGDPB110:~$ users
teja
teja@LAPTOP-FGDPB110:~$ who
teja      pts/1                2026-08-08 05:53
teja@LAPTOP-FGDPB110:~$ w
 05:55:54 up 2 min,  1 user,  load average: 0.01, 0.02, 0.00
USER      TTY      FROM              LOGIN@   IDLE   JCPU   PCPU   WHAT
teja      pts/1    -                  05:53    2:23   0.08s  0.06s  -bash
teja@LAPTOP-FGDPB110:~$ |

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```

teja@LAPTOP-FGDPB110:~$ groups
teja adm dialout cdrom floppy sudo audio dip video plugdev users netdev
teja@LAPTOP-FGDPB110:~$ env
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
NAME=LAPTOP-FGDPB110
PWD=/home/teja
LOGNAME=teja
HOME=/home/teja
LANG=C.UTF-8
WSL_INTEROP=/run/wsl/334_interop
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=00;33:so=01;35:do=01;35:bd=00;33;01:cd=00;33;01:or=00;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;
44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31
:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz
=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio
=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.
gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35
:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mpdv=0
1;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;3
5:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;
36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*.x=00;90:*.bak=00;90:*.
crdownload=00;90:*.dpg=dist=00;90:*.dpg-new=00;90:*.dpg-old=00;90:*.dpg-tmp=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.rpmnew=00;90:*.rpm
orig=00;90:*.rpmsave=00;90:*.swp=00;90:*.tmp=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:
WAYLAND_DISPLAY=wayland-0
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=teja
DISPLAY=:0
SHLVL=1
XDG_RUNTIME_DIR=/run/user/1000
WSLENv=
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/flatpak/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/Common Files/Oracle/Java
/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/system32
/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files/nodejs:/
mnt/c/Program Files/Git/cmd:/mnt/c/Users/teja/.local/bin:/mnt/c/Users/teja/AppData/Local/Microsoft/WindowsApps:/mnt/c/jdk-25_windows-x64_bin.zip/jdk-25:/mnt
/c/jdk-25_windows-x64_bin:/mnt/c/jdk-25_windows-x64_bin/jdk-25/bin:/mnt/c/Users/teja/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/teja/AppData/
Roaming/npm:/mnt/c/Users/teja/AppData/Local/Kingsoft/WPS Office/12.1.0.27458/clitool:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus

```

```

_=/usr/bin/printenv
teja@LAPTOP-FGDPB110:~$ df
Filesystem      1K-blocks      Used    Available Use% Mounted on
none            3970372         0     3970372   0% /usr/lib/modules/6.18.33
.2-microsoft-standard-WSL2
none            3970372         4     3970368   1% /mnt/wsl
drivers          498303300 134295152 364008148 27% /usr/lib/wsl/drivers
/dev/sdd         1055762868 1995676 1000063720 1% /
none            3970372         36     3970336   1% /mnt/wslg
none            3970372         0     3970372   0% /usr/lib/wsl/lib
rootfs           3964616      2772     3961844   1% /init
none            3970372        516     3969856   1% /run
none            3970372         0     3970372   0% /run/lock
none            3970372         0     3970372   0% /run/shm
none            3970372         80     3970292   1% /mnt/wslg/versions.txt
none            3970372         80     3970292   1% /mnt/wslg/doc
C:\              498303300 134295152 364008148 27% /mnt/c
tmpfs            794072         20     794052   1% /run/user/1000
teja@LAPTOP-FGDPB110:~$ free
              total        used        free      shared  buff/cache   avail
Mem:           7940748      553896      6686120          3568       855128       738
6852
Swap:          2097152           0      2097152
teja@LAPTOP-FGDPB110:~$ ps
  PID TTY          TIME CMD
   335 pts/0    00:00:00 bash
   549 pts/0    00:00:00 ps
teja@LAPTOP-FGDPB110:~$ top
top - 05:58:32 up 5 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.9 id,  0.1 wa,  0.0 hi,  0.0 si,  0.
MiB Mem :   7754.6 total,   6523.6 free,    546.4 used,    835.4 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,     0.0 used.   7208.3 avail Mem

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MiB Swap:   2048.0 total,   2048.0 free,     0.0 used.   7221.3 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
386	teja	20	0	20324	11576	9456	S	2.4	0.1	0:00.35	systemd
1	root	20	0	21876	13100	9736	S	0.0	0.2	0:01.71	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.04	init-systemd(Ub
6	root	20	0	3196	2132	2008	S	0.0	0.0	0:00.00	init
61	root	19	-1	50448	16196	15128	S	0.0	0.2	0:00.36	systemd-journal
108	root	20	0	25284	6672	5172	S	0.0	0.1	0:00.43	systemd-udevd
126	systemd+	20	0	21344	13216	11092	S	0.0	0.2	0:00.26	systemd-resolve
127	systemd+	20	0	91036	7984	7016	S	0.0	0.1	0:00.18	systemd-timesyn
188	root	20	0	4244	2680	2432	S	0.0	0.0	0:00.01	cron
191	message+	20	0	9644	5244	4540	S	0.0	0.1	0:00.14	dbus-daemon
198	root	20	0	17980	8884	7848	S	0.0	0.1	0:00.23	systemd-logind
201	root	20	0	1756628	13776	11444	S	0.0	0.2	0:00.23	wsl-pro-service
217	syslog	20	0	222516	5924	4640	S	0.0	0.1	0:00.15	rsyslogd
221	root	20	0	3124	1968	1836	S	0.0	0.0	0:00.01	agetty
231	root	20	0	107036	23216	13760	S	0.0	0.3	0:00.33	unattended-upgr
333	root	20	0	3184	1108	980	S	0.0	0.0	0:00.00	SessionLeader
334	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.04	Relay(335)
335	teja	20	0	6080	5404	3688	S	0.0	0.1	0:00.14	bash
336	root	20	0	6696	4652	3876	S	0.0	0.1	0:00.02	login
387	teja	20	0	21168	3596	1868	S	0.0	0.0	0:00.00	(sd-pam)
416	teja	20	0	6080	5348	3676	S	0.0	0.1	0:00.06	bash
550	teja	20	0	9300	5684	3496	R	0.0	0.1	0:00.03	top

```

Mem:           7940748      517752      6777396          3564      800208      742
Swap:         2097152          0      2097152

```

```
teja@LAPTOP-FGDPB110:~$ ps
```

```

PID TTY          TIME CMD
 463 pts/0      00:00:00 bash
 477 pts/0      00:00:00 ps

```

```
teja@LAPTOP-FGDPB110:~$ top
```

```

top - 06:01:23 up 7 min,  1 user,  load average: 0.09, 0.04, 0.01
Tasks:  22 total,   1 running,  21 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,   0.0 sy,   0.0 ni,100.0 id,   0.0 wa,   0.0 hi,   0.0 si,   0.
MiB Mem :  7754.6 total,  6617.9 free,   506.2 used,   781.5 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.  7248.4 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+
1	root	20	0	21848	13128	9796	S	0.0	0.2	0:01.68
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.03
6	root	20	0	3196	2124	2008	S	0.0	0.0	0:00.01
61	root	19	-1	42172	16200	15128	S	0.0	0.2	0:00.30
112	root	20	0	25020	6408	5148	S	0.0	0.1	0:00.38
126	systemd+	20	0	21344	13232	11108	S	0.0	0.2	0:00.24
127	systemd+	20	0	91036	7984	7024	S	0.0	0.1	0:00.16
190	root	20	0	4244	2688	2432	S	0.0	0.0	0:00.02
191	message+	20	0	9652	5364	4660	S	0.0	0.1	0:00.12
198	root	20	0	17976	8820	7780	S	0.0	0.1	0:00.14
200	root	20	0	1756628	13716	11444	S	0.0	0.2	0:00.18
208	syslog	20	0	222516	5672	4624	S	0.0	0.1	0:00.12
216	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.01
222	root	20	0	107036	23092	13636	S	0.0	0.3	0:00.31
330	root	20	0	6700	4664	3884	S	0.0	0.1	0:00.02
377	teja	20	0	20324	11516	9392	S	0.0	0.1	0:00.25
380	teja	20	0	21164	3572	1848	S	0.0	0.0	0:00.00
404	teja	20	0	6080	5364	3696	S	0.0	0.1	0:00.05
459	root	20	0	3184	1120	980	S	0.0	0.0	0:00.00
461	root	20	0	3200	1256	1108	S	0.0	0.0	0:00.00
463	teja	20	0	6080	5368	3648	S	0.0	0.1	0:00.12
478	teja	20	0	9300	5672	3488	R	0.0	0.1	0:00.00

```

sudo apt install sysstat
teja@LAPTOP-FGDPB110:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:                39 bits physical, 48 bits virtual
Byte Order:                   Little Endian
CPU(s):                       12
On-line CPU(s) list:         0-11
Vendor ID:                    GenuineIntel
Model name:                   13th Gen Intel(R) Core(TM) i5-1334U
CPU family:                   6
Model:                        186
Thread(s) per core:          2
Core(s) per socket:          6
Socket(s):                    1
Stepping:                     3
BogoMIPS:                     4992.02
Flags:                        fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr ss
                              e sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable
                              nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2api
                              c movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowpre
                              fetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi
                              1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetby
                              1 xsave avx_vnni vnmi umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_cl
                              ear serialize ibt flush_lld arch_capabilities

Virtualization features:
Virtualization:               VT-x
Hypervisor vendor:            Microsoft
Virtualization type:          full
Caches (sum of all):
L1d:                           288 KiB (6 instances)
L1i:                           192 KiB (6 instances)
L2:                             7.5 MiB (6 instances)
L3:                             12 MiB (1 instance)
NUMA:
NUMA node(s):                  1
NUMA node0 CPU(s):             0-11
Vulnerabilities:
Gather data sampling:          Not affected
Ghostwrite:                    Not affected
Indirect target selection:     Not affected

```

Vmscape: Not affected

teja@LAPTOP-FGDPB110:~\$ history

```
1  cffftcvgvvvvvvvvvv
2  clear
3  sudo apt update
4  sudo apt install build-essential installing:
5  gcc --version
6  gcc process . c -o process
7  ./process
8  clear
9  mkdir OSSP
10 cd OSSP
11 git init
12 mkdir src include obj bin docs tests scripts assets
13 touch .gitignore README.md LICENSE Makefile
14 touch src/main.c
15 touch src/shell.c
16 touch src/parser.c
17 touch src/executor.c
18 touch src/builtin.c
19 touch include/shell.h
20 touch include/parser.h
21 touch include/executor.h
22 touch include/builtin.h
23 touch docs/design.md
24 touch docs/report.pdf
25 touch tests/test_parser.c
26 touch tests/test_executor.c
27 touch scripts/run.sh
28 touch assets/architecture.png
29 tree
30 sudo apt install tree
31 tree
32sudo apt install tree
33sudo apt install build-essential
34gcc --version
35nano process.c
36sudo apt install tree
37sudo apt install build-essential
38nano process.c
39
```

```
48 cal
49 hostname
50 uname -a
51 exit
52 sudo apt update
53 sudo apt upgrade
54 sudo apt install gcc
55 gcc --version
56 nano process.c
57 gcc process.c -o process
58 ./process
59 ls
60 pwd
61 date
62 cal
63 whoami
64 hostname
65 uname
66 uname -a
67 ps
68 top
69 history
70 Ctrl + O
71 mkdir Week3
72 cd Week3
73 nano task1.c
74 Ctrl + O~
75 hello
76 mkdir OSSP
77 cd OSSP
78 mkdir src include obj bin docs tests scripts assets
79 mkdir OSSP
80 cd OSSP
81 mkdir src include obj bin docs tests scripts assets
82 ls
83 ls -l
84 find . -maxdepth 3
85 touch .gitignore README.md LICENSE Makefile
86 touch src/main.c src/shell.c src/parser.c src/executor.c src/builtin.c
87 touch include/shell.h include/parser.h include/executor.h include/builtin.h
88 touch docs/design.md docs/report.pdf
```

```
94 git status
95 git add .
96 git status
97 git commit -m "Initial OSSP project structure"
98 git status
99 nothing to commit, working tree clean
100 nano fork_example.c
101 gcc fork_example.c -o fork_example
102 ./fork_example
103 ls
104 pwd
105 data
106 whoami
107 uname
108 uname -r
109 id
110 tty
111 uptime
112 users
113 who
114 w
115 groups
116 env
117 printenv
118 df
119 free
120 ps
121 top
122 df
123 free
124 ps
125 top
126 free
127 ps
128 top
129 vmstatvmsat
130 vmstat
131 iostat
132 lscpu
133 history
teja@LAPTOP-FGDPB110:~$ |
```

```
127 ps
128 top
129 vmstatvmsat
130 vmstat
131 iostat
132 lscpu
133 history
teja@LAPTOP-FGDPB110:~$ clear|
```

```
teja@LAPTOP-FGDPB110:~$ echo hello
hello
teja@LAPTOP-FGDPB110:~$ bc
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software Foundation, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type `warranty'.
factor
0
factor 12
(standard_in) 2: syntax error
yes
0
sleep
0
|
```